



Dynamics of biofilm formation by *Listeria monocytogenes* on stainless steel under mono-species and mixed-culture simulated fish processing conditions and chemical disinfection challenges



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ABSTRACT

The progressive ability of a six-strains *L. monocytogenes* cocktail to form biofilm on stainless steel (SS), under fish-processing simulated conditions, was investigated, together with the biocide tolerance of the developed sessile communities. To do this, the pathogenic bacteria were left to form biofilms on SS coupons incubated at 15 °C, for up to 240 h, in periodically renewable model fish juice substrate, prepared by aquatic extraction of sea bream flesh, under both mono-species and mixed-culture conditions. In the latter case, *L. monocytogenes* cells were left to produce biofilms together with either a five-strains cocktail of four *Pseudomonas* species (*fragi*, *savastanoi*, *putida* and *fluorescens*), or whole fish indigenous microflora. The biofilm populations of *L. monocytogenes*, *Pseudomonas* spp., Enterobacteriaceae, H₂S producing and aerobic plate count (APC) bacteria, both before and after disinfection, were enumerated by selective agar plating, following their removal from surfaces through bead vortexing. Scanning electron microscopy was also applied to monitor biofilm formation dynamics and anti-biofilm biocidal actions. Results revealed the clear dominance of *Pseudomonas* spp. bacteria in all the mixed-culture sessile communities throughout the whole incubation period, with the in parallel sole presence of *L. monocytogenes* cells to further increase (ca. 10-fold) their sessile growth. With respect to *L. monocytogenes* and under mono-species conditions, its maximum biofilm population (ca. 6 log CFU/cm²) was reached at 192 h of incubation, whereas when solely *Pseudomonas* spp. cells were also present, its biofilm formation was either slightly hindered or favored, depending on the incubation day. However, when all the fish indigenous microflora was present, biofilm formation by the pathogen was greatly hampered and never exceeded 3 log CFU/cm², while under the same conditions, APC biofilm counts had already surpassed 7 log CFU/cm² by the end of the first 96 h of incubation. All here tested disinfection treatments, composed of two common food industry biocides gradually applied for 15 to 30 min, were insufficient against *L. monocytogenes* mono-species biofilm communities, with the resistance of the latter to significantly increase from the 3rd to 7th day of incubation. However, all these treatments resulted in no detectable *L. monocytogenes* cells upon their application against the mixed-culture sessile communities also containing the fish indigenous microflora, something probably associated with the low attached population level of these pathogenic cells before disinfection (< 10² CFU/cm²) under such mixed-culture conditions. Taken together, all these results expand our knowledge on both the population dynamics and resistance of *L. monocytogenes* biofilm cells under conditions resembling those encountered within the seafood industry and should be considered upon designing and applying effective anti-biofilm strategies.

1. Introduction

L. monocytogenes is a significant foodborne bacterial pathogen provoking listeriosis, a relatively rare but still life-threatening disease in immunocompromised individuals (Gandhi and Chikindas, 2007). In Europe, 2161 confirmed human listeriosis cases and 210 deaths were recorded in 2014 (EFSA and ECDC, 2015). Alarming, this year, the

hospitalization and case-fatality ratios were 98.9% and 15%, respectively. *L. monocytogenes* can adhere to and produce biofilms on many surfaces encountered within the food industry (Valderrama and Cutter, 2013), resulting in potential continuous contamination of the products (Hansen and Vogel, 2011; Tresse et al., 2007). These multicellular sessile microbial communities are embedded in hydrated extracellular polymeric matrixes and their formation is more and more considered as

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a universal microbial adaptation mechanism against various stresses and antimicrobials, like those encountered within food processing (Bridier et al., 2015). To this direction, numerous studies have shown that *L. monocytogenes* strains of different origins and serotypes, are able to produce biofilms on a variety of surfaces, both biotic and abiotic, depending on the strain, surface and culture conditions (da Silva and De Martinis, 2013; Harvey et al., 2007; Mosquera-Fernández et al., 2014; Nilsson et al., 2011; Poimenidou et al., 2016). Indeed, the great persistence this pathogen may display in food processing plants upon its enclosure in biofilm structures, even for years, is believed to seriously contribute to its transmission and be the reason behind several human listeriosis outbreaks (Carpentier and Cerf, 2011; Ferreira et al., 2014; Nowak et al., 2017).

Seafood consumption is worldwide an integral part of a healthy diet. However, this is not without risk, with seafood commodities commonly implicated in food-borne disease outbreaks, caused by a variety of bacteria, viruses and parasites (Iwamoto et al., 2010). Although *L. monocytogenes* may be present in many foods, ready-to-eat (RTE) ones, such as marinated or cold smoked fishes, require the greatest attention. This is indeed included in the common seafood bacterial pathogens, with the proportion of positive for this bacterium samples in RTE foods at European Union retail markets, in 2014, to be highest in fish products (mainly smoked fish) (EFSA and ECDC, 2015). Sea bream (*Sparus aurata*) is one of the most important cultured fish species in the Mediterranean region and especially Greece, which is the leading producer in the world with 44.3% of the total production (FAO, 2011–2017). Alarming, biofilm formation has been problematic in the aquaculture industry from farm (preharvest) through the processing plant (post-harvest) (Mizan et al., 2015; Rajkowski, 2009). Although the prevalence of *L. monocytogenes* in European raw fish could be considered low, with an overall one of 3% to have been described (Davies et al., 2001), the fact that these bacteria are psychrotrophic and can also grow under modified atmospheres and/or high salt content is of concern (Provincial et al., 2013). The risk of infection increases considering that fish and fishery products may also sometimes be eaten undercooked.

Given this serious risk, previous research has focused on the ecology and prevalence of *L. monocytogenes* in fish processing environments (Hoffman et al., 2003; Leong et al., 2015; Nakamura et al., 2013; Thimothe et al., 2002). In such a study, all raw RTE seafood isolates of *L. monocytogenes* tested ($n = 61$) could form biofilms to various degrees (Takahashi et al., 2009). However, most of the studies on the ability of *L. monocytogenes* to colonize abiotic surfaces have been conducted under monoculture conditions, and with the use of laboratory (synthetic) growth media to support growth. Nevertheless, this does not sufficiently represent the real situation encountered in food processing environments, where *L. monocytogenes* may be present on surfaces with many other bacteria forming multi-species biofilms (Dzieciol et al., 2016; Liu et al., 2016). Under such conditions, interspecies interactions are unavoidable and may affect both the dynamics of biofilm formation by each individual species and their resistance (Giaouris et al., 2015). Thus, the potential protection of pathogenic cells by resident surface flora inside mixed-culture biofilms when subjected to disinfection is surely a matter of concern (Sanchez-Vizuet et al., 2015). An example of this is *L. monocytogenes* surviving in biofilms formed by *Pseudomonas* (Puga et al., 2016). These latter are Gram-negative aerobic bacteria which are commonly found as the main components of surface flora isolated from food processing environments following regular cleaning and disinfection (Fagerlund et al., 2017; Guobjörnsdóttir et al., 2005; Langsrud et al., 2016; Liu et al., 2016) and have been also described as good biofilm producers (Mann and Wozniak, 2012).

In food industry, conditioning films are quickly developed on various surfaces following the absorption of different types of food residues onto them (Whitehead et al., 2010). Such films can influence both microbial biofilm formation on these surfaces and in parallel the subsequent resistance of surface-attached cells to various stresses, including cleaning and disinfection, causing the failure of sanitization

processes and leading to persistence. For instance, the behavior of sessile *L. monocytogenes* cells has been previously found to be affected by the type of food soil (Gram et al., 2007; Kuda et al., 2015; Overney et al., 2016; Takahashi et al., 2011). It is therefore recommended, whenever possible, to study biofilm formation and/or resistance of sessile cells by using, in the laboratory, food soils or media simulating the ones likely to be encountered by microorganisms in real food processing environments to better understand and predict their sessile behavior.

Considering all the above, the objective of the current study was to assess the progressive ability of a six-strains *L. monocytogenes* cocktail to form biofilm on stainless steel (SS), under fish-processing simulated mono-species or mixed-culture conditions, together with the biocide tolerance of the developed sessile communities against two common food industry chemical disinfectants. In the case of mixed-culture conditions, pathogenic cells were left to produce biofilms together with either a five-strains cocktail of four *Pseudomonas* species (*fragi*, *savastanoi*, *putida* and *fluorescens*), or whole fish indigenous microflora. Model fish juice substrate was always used to support sessile development.

2. Materials and methods

2.1. Bacterial strains, media and preparation of the inocula

Six *L. monocytogenes* and five *Pseudomonas* spp. strains were used in this study (Table 1). Regarding the *L. monocytogenes* strains, two were of clinical origin (FMCC_B124; FMCC_B125), two had been isolated from foods (FMCC_B127; FMCC_B128), and two from food processing environments (FMCC_B131; FMCC_B157). Regarding the *Pseudomonas* spp. strains, one belonged to *P. fragi* species (FMCC_B209), one to *P. savastanoi* species (ATCC13526), two to *P. putida* species (ATCC12633; FMCC_B74), and one to *P. fluorescens* species (FMCC_B29). All strains, apart from ATCC13526 and ATCC12633, were kindly provided by Prof. G.-J. Nychas from the Agricultural University of Athens (Athens, Greece).

Before utilization, all bacteria were stored frozen at -80°C in bead vials (Protect; Technical Service Consultants, Ltd., Heywood, Lancashire, United Kingdom) and were then resuscitated by adding one bead for each strain to 10 ml of Tryptone Soy Broth (TSB; Lab M; International Diagnostics Group Plc, Bury, Lancashire, UK) and incubating for 24 h at either 37°C (*L. monocytogenes* strains) or 25°C (*Pseudomonas* spp. strains) (precultures). These temperatures were chosen to be closely to the optimum for each genus to quickly and successfully resuscitate the bacteria following their deep-frozen storage. Working cultures were prepared by adding a 10- μl aliquot of each preculture to 10 ml of TSB and incubating for another 24 h at the appropriate temperature mentioned above. Cells from final (stationary phase) working cultures were harvested by centrifugation ($2500 \times g$ for 10 min at room temperature), washed twice and re-suspended in sterile saline solution (0.85% w/v NaCl). Finally, the bacterial suspensions of the six *L. monocytogenes* and/or the five *Pseudomonas* spp. strains were combined and further diluted in sterile saline solution to yield inocula of approximately 10^{7-8} CFU/ml, to be used for the subsequent biofilm formation assays. It should be noted that care was taken in order all these three inocula to contain approximately the same number of cells for each strain.

2.2. Preparation of fish juice broth (FJB) model substrate

Sterile FJB model substrate was prepared by the extraction method described by Parlapani et al. (2017). Briefly, 3 kg of fresh fish farm breams were purchased from a fish shop in Volos (Greece) and transported, in ice, to the laboratory within 1 h of purchase. There, 250 g of their flesh were aseptically cut in small pieces and homogenized with 250 ml of sterile deionized water, for 2 min, using a Stomacher (Bug

Table 1
Bacterial strains used in this study and their characteristics.

Bacterial species	Strain ^a	Features	Origin	Reference
<i>Listeria</i> spp. strains				
<i>L. monocytogenes</i>	FMCC_B124, ATCC13932, NCTC10527	Serotype 4b	Spinal fluid of child with meningitis, Germany	Tominaga, 2006
<i>L. monocytogenes</i>	FMCC_B125, ScottA	Serotype 4b, lineage I, epidemic strain	Human	Schlech et al., 1983
<i>L. monocytogenes</i>	FMCC_B127, cretan collection: 21,075		Chicken salad	Kostaki et al., 2012
<i>L. monocytogenes</i>	FMCC_B128, cretan collection: 21,085		Soft cheese	Argyri et al., 2013
<i>L. monocytogenes</i>	FMCC_B131, cretan collection: 214,112		Conveyor belt of ready-to-eat (RTE) frozen foods	Argyri et al., 2013
<i>L. monocytogenes</i>	FMCC_B157, DSA23UD		Food processing plant	Cocolin et al., 2005
<i>Pseudomonas</i> spp. strains				
<i>P. fragi</i>	FMCC_B209, ATCC4973, DSM3456	Type strain		Ercolini et al., 2007
<i>P. savastanoi</i>	ATCC13526		Oleander	Azad and Cooksey, 1995
<i>P. putida</i>	ATCC12633, NCTC10936, DSM291	Type strain, biotype A		Chang and Yen, 1985
<i>P. putida</i>	FMCC_B74		Bream fillet	This study
<i>P. fluorescens</i>	FMCC_B29		Aquaculture sea bream	This study

ATCC, American Type Culture Collection; NCTC, National Collection of Type Cultures; DSA, Dipartimento di Scienze degli Alimenti, Udine, Italy; DSMZ, Deutsche Sammlung von Mikroorganismen und Zellkulturen (German Collection of Microorganisms and Cell Cultures).

^a FMCC, Food Microbiology Culture Collection, Laboratory of Microbiology and Biotechnology of Foods, Agricultural University of Athens, Greece;

Mixer; Interscience, London, UK). This procedure was appropriately repeated to collect the necessary volume of juice required for all the experiments described below. The juice was separated from solids through tulle and was boiled for 5 min. Then, it was left for 15 min at RT and subsequently filtered using Whatman® No.1 cellulose filter papers (Whatman Inc.; Clifton, NJ, USA). Afterwards, 0.10 M of phosphate buffer (0.056 M KH₂PO₄ and 0.044 M K₂HPO₄) was added and pH was adjusted to 6.5 (with NaOH or HCl). Finally, the FJB was boiled for 30 min, before aseptically placed, in properly weighed portions (100 ml), in sterile Durhan bottles and stored frozen at – 20 °C until use. It should be noted that the 1st short boiling step (for 5 min) was done to facilitate the subsequent filtration, whereas the 2nd long boiling step (for 30 min) was sufficient to sterilize the broth (as verified by plate counting). When raw FJB was used to support biofilm development (Section 2.3.3), the two boiling steps described above were omitted and care was taken to avoid as much as possible any exogenous contamination of the samples. Immediately after its preparation, the indigenous bacterial load of the freshly prepared raw FJB (0 h), concerning the counts for *Pseudomonas* spp., Enterobacteriaceae, H₂S producing bacteria and aerobic plate counts (APC), was quantified by serial decimal dilutions and selective agar plating, using media and incubation conditions described below (Section 2.5).

2.3. Biofilm formation

2.3.1. Abiotic surfaces

Stainless steel (SS) coupons (3 × 1 × 0.1 cm, type AISI-304; Halvourgiki Inc.; Athens, Greece) were the abiotic substrates used to support biofilm development. This SS type is commonly used in the food processing industry for fabricating equipment due to its physico-chemical stability and high resistance to corrosion. Before use, coupons were thoroughly cleaned and sterilized following the procedure described by Kostaki et al. (2012).

2.3.2. Biofilm formation by *L. monocytogenes* and/or *Pseudomonas* spp.

L. monocytogenes and/or *Pseudomonas* spp. bacteria were left to form biofilms on SS coupons, incubated in sterile FJB, under either mono- or dual-genus conditions (i.e. either strains of one single genus or strains of both genera together), following the protocol described by Kostaki et al. (2012), with some adaptations. Briefly, 5 ml of each saline inoculum (mono-/dual-genus; ca. 10^{7–8} CFU/ml) was added into each glass test tube containing a sterile SS coupon and was incubated at

15 °C for 3 h, to allow bacteria to attach to the coupons. Afterwards, each coupon was aseptically removed from the glass test tube and was then rinsed, by pipetting 6 ml of saline solution (i.e. 3 ml on its side), to remove the loosely attached cells. Each coupon was thereupon introduced into a new sterile glass test tube containing 5 ml of sterile FJB and statistically incubated at 15 °C for up to 240 h, to allow attached bacteria to grow and form biofilms on it. Growth medium was renewed every 48 h. Before the addition of new sterile medium, loosely attached cells were removed by rinsing (as described above).

2.3.3. Biofilm formation by fish indigenous microflora either alone or together with *L. monocytogenes*

When raw FJB was the medium used to support biofilm development by fish indigenous microflora, either alone or together with *L. monocytogenes*, each sterile SS coupon was directly incubated in 5 ml of: (i) either raw FJB, or (ii) raw FJB also artificially inoculated (at 0 h) with the six *L. monocytogenes* strains inoculum (at an initial concentration of ca. 10^{2–3} CFU/ml). Incubation was again carried out at 15 °C for up to 240 h, to allow endogenous microorganisms of FJB alone (in the first case), or together with *L. monocytogenes* bacteria (in the second case), to attach and form mixed-culture biofilms on SS. Growth medium was again renewed every 48 h. Before the addition of new raw medium, loosely attached cells were removed by rinsing (as described above).

2.4. Disinfection of mono-species and mixed-culture *L. monocytogenes* biofilms

2.4.1. Chemical disinfectants

Two common food industry chemical disinfectants were tested: (i) Hypofoam VF6, and (ii) Divosan SU 319/VT8w. Both were manufactured by Diversey Hellas S.A. (SealedAir; Marousi, Athens, Greece) and their desired working solutions (10% and 1%, respectively) were prepared in situ the day of application in tap water. Both disinfectants were kindly provided from DIAS Aquaculture SA (Athens, Greece), a company operating fish-farming units, hatcheries, and fish processing plants and which routinely applies them, in those concentrations described above, in its daily sanitation program.

2.4.2. Disinfection procedure

L. monocytogenes mono-species biofilms (Section 2.3.2) were submitted to disinfection on both the 3rd and 7th day of incubation (i.e. 72

and 168 h, respectively), while mixed-culture biofilm communities, composed of fish indigenous microflora, either alone or together with *L. monocytogenes* (Section 2.3.3), were submitted to disinfection solely on the 7th day of incubation. To do this, the day of disinfection, each SS coupon, carrying biofilm bacteria onto it, was carefully removed from the glass test tube using sterile forceps and was thereafter rinsed, by pipetting 6 ml of saline solution (i.e. 3 ml on its side), to remove the loosely attached cells. After being rinsed, the coupon was individually introduced into a new glass test tube containing 5 ml of the disinfectant solution. Four different disinfection treatments were always evaluated: application of Hypofoam for 15 min (i), application of Hypofoam for 15 min and subsequent application of Divosan for 5 (ii), 10 (iii) or 15 min (iv). Hypofoam was applied at 37 °C, whereas Divosan at 20 °C, in accordance with the manufacturer's instructions and HACCP industry guidelines.

2.5. Recovery and quantification of biofilm cells

The periodic quantification of biofilm bacteria on SS coupons at selected time intervals through incubation (up to 240 h; Sections 2.3.2 and 2.3.3), together with those remaining viable following each disinfection treatment (Section 2.4.2), was performed by using the bead vortexing and agar plating method described by Kostaki et al. (2012), with some minor adaptations. Briefly, at the time of sampling, each SS coupon was aseptically removed from the glass test tube and was thereafter rinsed, by pipetting 6 ml of saline solution (i.e. 3 ml on its side), to remove either the loosely attached cells or the disinfectants' residues. Biofilm bacteria were subsequently detached by strong 1-min vortexing of each coupon in plastic centrifuge tube containing 6 ml of saline solution (0.85% w/v NaCl) and 35 sterile glass beads (diameter 1.5 mm). The recovered bacteria were subsequently enumerated by duplicate agar plating using the appropriate selective growth medium and incubation conditions described below, after ten-fold serial dilutions in sterile saline solution. Thus, *L. monocytogenes* bacteria were enumerated by spread-plating (0.1 ml) on Polymyxin, Acriflavine, Lithium chloride, Ceftazidime, Aesculin, Mannitol (PALCAM) Agar, *Pseudomonas* spp. bacteria were enumerated by spread-plating (0.1 ml) on Cephalothin, Fucidin, Ceftrimide (CFC) Agar, Enterobacteriaceae by pour-plating (1 ml) in Violet Red Bile Glucose Agar (VRBGA), *H₂S* producing bacteria by pour-plating (1 ml) and counting only black colonies in Iron Agar (IA), and finally, total aerobic plate count (APC) was enumerated by pour-plating (1 ml) and counting all colonies in IA. PALCAM agar and VRBGA plates were incubated at 37 °C for 24 h, while CFC agar and IA plates at 25 °C for 48 and 72 h, respectively. It

should be noted that Tryptone Soy Agar (TSA) plates were also inoculated in the cases of quantification of both mono- and dual-genus biofilm communities, just to verify that the two selective growth media applied (i.e. PALCAM and CFC) do not pose any significant obstacle to the growth of target bacteria (data not shown). Preliminary experiments had also verified that *L. monocytogenes* cells cannot grow on CFC agar, while *Pseudomonas* spp. cells cannot grow on PALMAM agar. All growth media were purchased by Lab M, except IA which was in situ prepared by its individual ingredients according to Gram et al. (1987).

2.6. SEM observation of mono-genus biofilm communities

SEM was employed to visualize mono-genus biofilm communities (composed of either *L. monocytogenes* or *Pseudomonas* spp. strains) grown in periodically renewable sterile FJB, before and after disinfection treatments. To do this, following the rinsing of the SS coupons with saline solution, these were fixed in 2.5% glutaraldehyde in 0.1 M sodium cacodylate solution for 12 h. After this fixation step, specimens were rinsed at both sides with 0.1 M sodium cacodylate solution and dehydrated by immersing in a graded alcohol series (30, 50, 60, 70, 90, 95 and 100%). The specimens were finally covered with a thin layer of gold using a sputter coater (Bal-tec SCD 004), before their examination under a scanning electron microscope (Cambridge Stereoscan 240).

2.7. Statistical analyses

Each experiment included three technical replicates (SS coupons) and was twice repeated using independent bacterial cultures. Biofilm counts on SS (CFU/cm²) were transformed to logarithms before means and standard deviations were calculated. The data obtained from the disinfection experiments were also analyzed by factorial analysis of variance (ANOVA), to check for any significant effects of microorganisms and/or disinfection treatment on the log reductions of biofilm cells (log CFU/cm²), using the statistical software STATISTICA (StatSoft Inc. v.12; Tulsa, OK 74104, USA). Least square means were separated by Fisher's least significant difference (LSD) test. All differences are reported at a significance level of 0.05.

3. Results

3.1. Dynamics of biofilm formation by *L. monocytogenes* and *Pseudomonas* spp.

The dynamics of biofilm formation by *L. monocytogenes* and

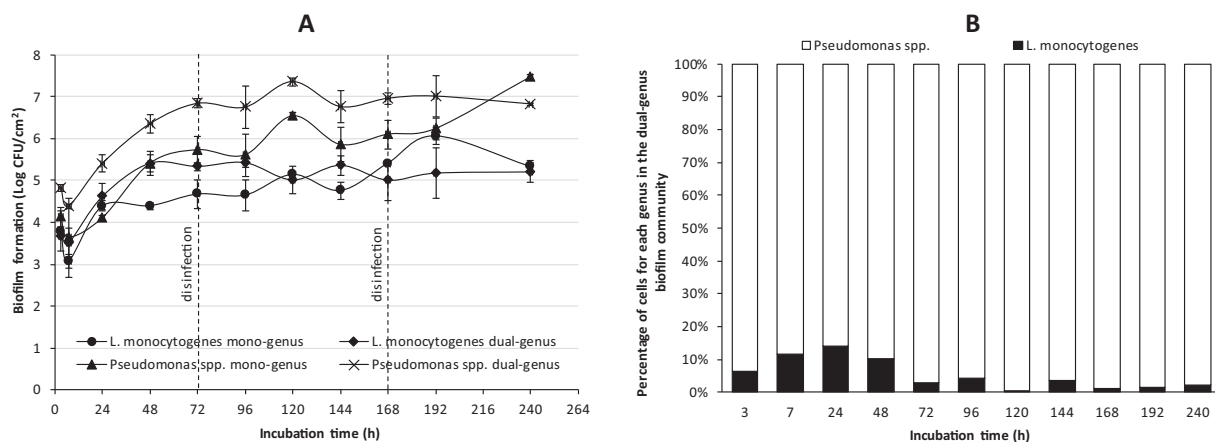


Fig. 1. (A) Dynamics of biofilm formation (log CFU/cm²) by *L. monocytogenes* (●, ◆) and *Pseudomonas* spp. (▲, x) strains, under either mono-genus (●, ▲) or dual-genus (◆, x) conditions. The points represent mean values ± standard deviations. The vertical dashed lines indicate the times (i.e. 72 and 168 h) at which the *L. monocytogenes* mono-species biofilm communities were submitted to disinfection treatments. (B) Percentages of *L. monocytogenes* (filled bars) and *Pseudomonas* spp. (open bars) cells in the dual-genus biofilm communities (data extracted from subfigure A) throughout the incubation time. At all cases biofilms were left to be formed on SS coupons incubated at 15 °C, for a total period of up to 240 h (i.e. 10 d), in periodically (i.e. every 48 h) renewable sterile FJB, following an initial attachment step, at 15 °C for 3 h, in saline bacterial suspensions.

Pseudomonas spp. strains, grown under either mono- or dual-genus conditions, are presented in Fig. 1A, while the percentages of cells for each genus in the dual-genus biofilm communities throughout the incubation time, are better shown in Fig. 1B (last data extracted from subfigure Fig. 1A). Firstly, it can be observed that under the dual-genus conditions, the simultaneous presence of *L. monocytogenes* cells significantly favored biofilm formation by the *Pseudomonas* spp. bacteria. Thus, apart from the last day of incubation (240 h), *Pseudomonas* spp. biofilm counts were almost 1 log higher (i.e. ca. 10 times more) under the dual-genus compared to the mono-genus conditions. Secondly, under the same conditions, the simultaneous presence of *Pseudomonas* spp. cells either favored or hindered biofilm formation by *L. monocytogenes*, depending on the incubation day. However, both in the middle and by the end of incubation (i.e. at 120 and 240 h, respectively), no significant differences in *L. monocytogenes* biofilm counts were observed between the mono- and dual-genus conditions. Thirdly, *Pseudomonas* spp. bacteria dominated in an overwhelming way in the dual-genus biofilm communities throughout the whole incubation period (from 3 to 240 h), with their percentages ranging from 86 to 99.6% (Fig. 1B). Fourthly, under the mono-genus conditions, *Pseudomonas* spp. biofilm counts were significantly higher (more than ca. 1 log) from those of *L. monocytogenes* from the 48 h of incubation and thereafter (apart from 192 h). Remarkably, it was under these conditions where the maximum biofilm counts were observed for both bacterial genera (6.07 ± 0.21 and 7.47 ± 0.06 log CFU/cm² for *L. monocytogenes* and *Pseudomonas* spp., at 192 and 240 h, respectively). Irrespectively of bacteria and conditions tested (mono-/dual-genus), a rapid increase in the number of attached/biofilm cells occurred the first 48 h of incubation. Some representative SEM images of *L. monocytogenes* and *Pseudomonas* spp. attached/biofilm cells to SS coupons under the mono-genus conditions, at different incubation times, are shown in Figs. 2 and 3, respectively. The progressive growth of single attached bacteria to microcolonies and finally to three-dimensional biofilm structures containing rich amounts of extracellular polymeric

substances (EPS), is clear for both bacterial genera.

3.2. Dynamics of biofilm formation by fish indigenous microflora either alone or together with *L. monocytogenes*

The dynamics of biofilm formation by the main categories of the fish indigenous microflora here studied (i.e. *Pseudomonas* spp., Enterobacteriaceae, H₂S producing bacteria, and APC), either alone (i), or together with *L. monocytogenes* (ii), are shown in Fig. 4A and B, respectively. A rapid increase in the number of attached/biofilm cells is observed the first 96 h of incubation, with APC biofilm counts surpassing 7 log CFU/cm² by the end of this period, at both cases. *Pseudomonas* spp. bacteria were again found to dominate in both the mixed-culture biofilm communities, throughout the incubation time. The levels of planktonic populations (log CFU/ml) for the different microbial groups found in the raw FJB at 0 h are presented in Table 2. Thus, raw FJB was found to contain an APC of 3.77 ± 0.21 log CFU/ml at the beginning of incubation. Concerning the final numbers of APC biofilm populations found on SS coupons at the end of incubation (240 h), these were equal to 6.76 ± 0.68 and 7.71 ± 0.33 log CFU/cm², for the 1st and 2nd case, respectively. However, under these latter mixed-culture conditions, the maximum number of *L. monocytogenes* biofilm counts never exceeded the barrage of 3 logs, with the higher biofilm population for these bacteria to be equal to 2.73 ± 0.25 log CFU/cm² and observed the 4th day of incubation (96 h), whereupon a progressive decrease in their sessile population resulted in the level of 1.62 ± 0.40 log CFU/cm² at the end of incubation (240 h) (Fig. 4B).

3.3. Disinfection of *L. monocytogenes* mono-species biofilms

The log reductions (log CFU/cm²) of *L. monocytogenes* mono-species biofilm communities observed following the four different disinfection treatments (Section 2.4.2), applied at either 72 or 168 h (i.e. 3 and 7 d, respectively) of incubation left for biofilm formation on SS coupons

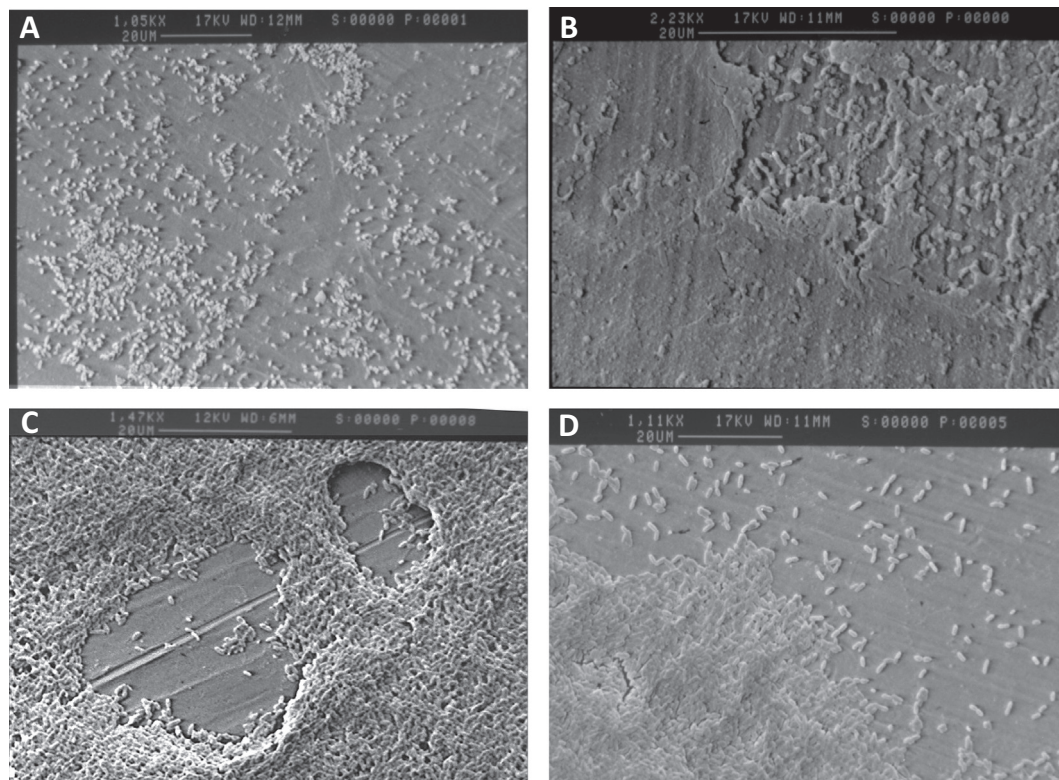


Fig. 2. SEM micrographs of *L. monocytogenes* cells on SS coupons incubated under mono-species conditions at 15 °C, for 7 h (A), 120 h (B), 168 h (C) and 240 h (D), in periodically (i.e. every 48 h) renewable sterile FJB, following an initial attachment step, at 15 °C for 3 h, in saline bacterial suspension.

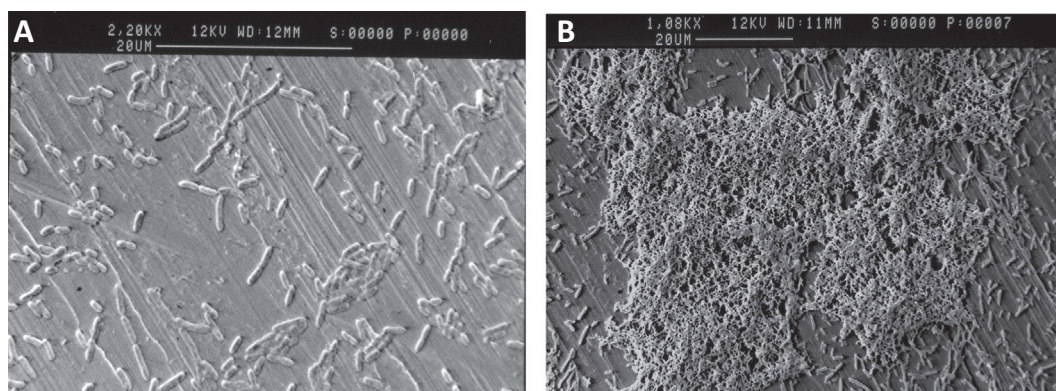


Fig. 3. SEM micrographs of *Pseudomonas* spp. cells on SS coupons incubated under mono-genus conditions at 15 °C, for 24 h (A) and 240 h (B), in periodically (i.e. every 48 h) renewable sterile FJB, following an initial attachment step, at 15 °C for 3 h, in saline bacterial suspension.

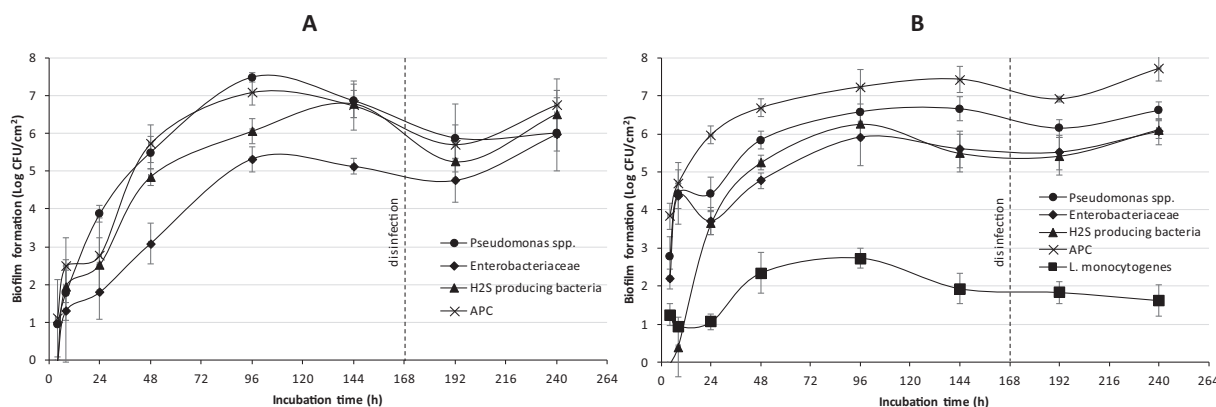


Fig. 4. Dynamics of biofilm formation (log CFU/cm²) by *Pseudomonas* spp. (●), Enterobacteriaceae (◆), H₂S producing (▲), APC (x) and *L. monocytogenes* (■) bacteria, under mixed-culture conditions. The points represent mean values $\pm$ standard deviations. The vertical dashed lines indicate the time (i.e. 168 h) at which the mixed-culture biofilm communities were submitted to disinfection treatments. At all cases biofilms were left to be formed on SS coupons incubated at 15 °C, for a total period of up to 240 h (i.e. 10 d), in periodically (i.e. every 48 h) renewable raw FJB (A); or raw FJB also artificially inoculated (at 0 h) with *L. monocytogenes* (at an initial concentration of 2.53 ± 0.16 log CFU/ml) (B).

Table 2

Planktonic populations (log CFU/ml) of microorganisms found in raw FJB at 0 h, together with their biofilm populations (log CFU/cm²) at 168 h. Biofilms were left to be formed on SS coupons incubated at 15 °C in periodically (i.e. every 48 h) renewable raw FJB, or raw FJB also artificially inoculated (at 0 h) with *L. monocytogenes*.

Microorganisms	Planktonic population in raw FJB at 0 h (log CFU/ml)	Biofilm population at 168 h (log CFU/cm ²)	
		Raw FJB	Raw FJB & <i>L. monocytogenes</i>
<i>Pseudomonas</i> spp.	3.14 $\pm$ 0.15	6.76 $\pm$ 0.23	6.56 $\pm$ 0.14
Enterobacteriaceae	2.08 $\pm$ 0.36	7.25 $\pm$ 0.35	6.03 $\pm$ 0.35
H ₂ S producing bacteria	2.42 $\pm$ 0.38	5.96 $\pm$ 1.62	5.82 $\pm$ 0.28
APC	3.77 $\pm$ 0.21	7.15 $\pm$ 0.55	7.55 $\pm$ 0.21
<i>L. monocytogenes</i> ^a	2.53 $\pm$ 0.16	–	1.89 $\pm$ 0.52

^a Artificially inoculated.

(Section 2.3.2), are shown in Fig. 5. Apart from applying Hypofoam alone (for 15 min), an increased resistance of the 7th day biofilms over those of the 3rd day, is observed for the other 3 treatments. Thus, the log reductions of *L. monocytogenes* biofilm cells at the latter case (3rd day) were always significantly higher than those of the former case (7th day). However, these log reductions never exceeded the barrage of 4 logs, even when a 30-min disinfection treatment was applied (i.e. each disinfectant for 15 min) against the 3 d sessile communities. It is worth mentioning that *L. monocytogenes* biofilm counts encountered on SS coupons before disinfection, at 72 and 168 h of incubation, were equal to 4.67 ± 0.34 and 5.41 ± 0.05 log CFU/cm², respectively (Fig. 1A). Two representative SEM images of *L. monocytogenes* biofilm cells

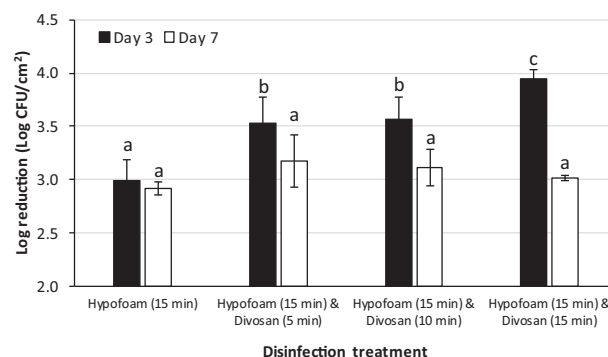


Fig. 5. Log reductions (log CFU/cm²) of *L. monocytogenes* biofilm cells following the four different disinfection treatments. Before disinfection, bacteria had been left form biofilms, under mono-species conditions, on SS coupons incubated at 15 °C, for either 72 (■) or 168 h (□) (i.e. 3 and 7 d, respectively), in periodically (i.e. every 48 h) renewable sterile FJB, following an initial attachment step, at 15 °C for 3 h, in saline bacterial suspension. The bars represent mean values $\pm$ standard deviations. Mean values sharing at least one common lowercase letter shown above the bars are not significantly different at a *P* value of 0.05.

following the combined successive action of Hypofoam and Divosan (both applied for 15 min) are shown in Fig. 6. The breakdown of biofilm structure (Fig. 6A), together with the destruction of the pathogenic sessile cells, found to present visible holes (Fig. 6B), are evident.

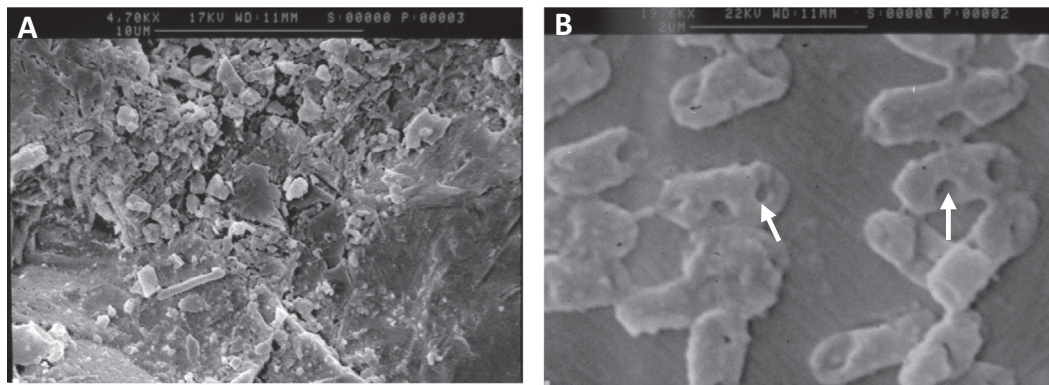


Fig. 6. SEM micrographs of *L. monocytogenes* biofilm cells following the combined successive action of Hypofoam and Divosan (both applied for 15 min) (bar scales 10 and 2 µm, for A and B, respectively). The arrows indicate holes created in cells due to biocides' action. Before disinfection, bacteria had been left form biofilms, under mono-species conditions, on SS coupons incubated at 15 °C for 72 h, in periodically (i.e. every 48 h) renewable sterile FJB, following an initial attachment step, at 15 °C for 3 h, in saline bacterial suspension.

3.4. Disinfection of mixed-culture biofilms composed of fish indigenous microflora either alone or together with *L. monocytogenes*

The log reductions ($\log \text{CFU}/\text{cm}^2$) of biofilm populations for the different microbial groups (*Pseudomonas* spp., Enterobacteriaceae, H_2S producing bacteria, and APC) of the fish indigenous microflora, either incubated alone (i), or together with *L. monocytogenes* (ii), are shown in Fig. 7A and B, respectively. Concerning the levels of their biofilm populations ($\log \text{CFU}/\text{cm}^2$) encountered on SS coupons before disinfection (at 168 h of incubation), these are, for both cases, presented in Table 2. Thus, no significant differences in these populations seem to exist between the 2 cases (i.e. incubation in raw FJB alone or FJB also artificially inoculated, at the beginning, with *L. monocytogenes*), except for the case of Enterobacteriaceae biofilm counts, which were significantly higher in the former compared to the latter case (7.25 ± 0.35 and $6.03 \pm 0.35 \log \text{CFU}/\text{cm}^2$, respectively) (Table 2). In the case of the solely presence of fish indigenous microflora, it is observed that the log reductions significantly increased by going from the application of Hypofoam alone for 15 min to the 30-min disinfection treatment (i.e. each disinfectant applied for 15 min) (Fig. 7A). Under such conditions, no significant differences existed between the two most harsh treatments, i.e. application of Hypofoam for 15 min in combination with either a 10 or 15 min application of Divosan. However, both these last treatments resulted in remaining viable biofilm cells which were always below the detection limits (DLs) of the plate counting methods (0.95 and $-0.05 \log \text{CFU}/\text{cm}^2$, for spread- and pour-plate methods,

respectively), apart from the combined application of Hypofoam for 15 min and Divosan for 10 min against APC. This increase in the efficiency of disinfection treatments by increasing their duration, without still having any significant difference between the two most harsh treatments, was also again evident against the total sessile aerobic fish microflora (APC) also containing the pathogenic cells (Fig. 7B). In addition, under such conditions, all four applied treatments resulted in undetectable *L. monocytogenes* cells following the disinfection. However, the fact that the sessile pathogenic population was below $2 \log \text{CFU}/\text{cm}^2$ before the disinfection (Table 2), should normally account for its disappearance following this. It is worth noting that under such mixed-culture conditions (spoilage and pathogenic bacteria together), for all the three main microbial groups tested (*Pseudomonas* spp., Enterobacteriaceae and H_2S producing bacteria), no significant differences were observed between the efficiency of the four different disinfection treatments, with even the mildest one (i.e., application of Hypofoam for 15 min) to cause always $> 4 \log$ reductions. This last observation reveals an effect of the incorporation of *L. monocytogenes* cells in the overall resistance of the mixed-culture biofilms developed by the fish indigenous microflora, which is especially interesting considering that pathogenic bacteria consisted only the ca. 0.0002% of the total sessile APC before disinfection (1.89 and $7.55 \log \text{CFU}/\text{cm}^2$, on average, respectively; Table 2).

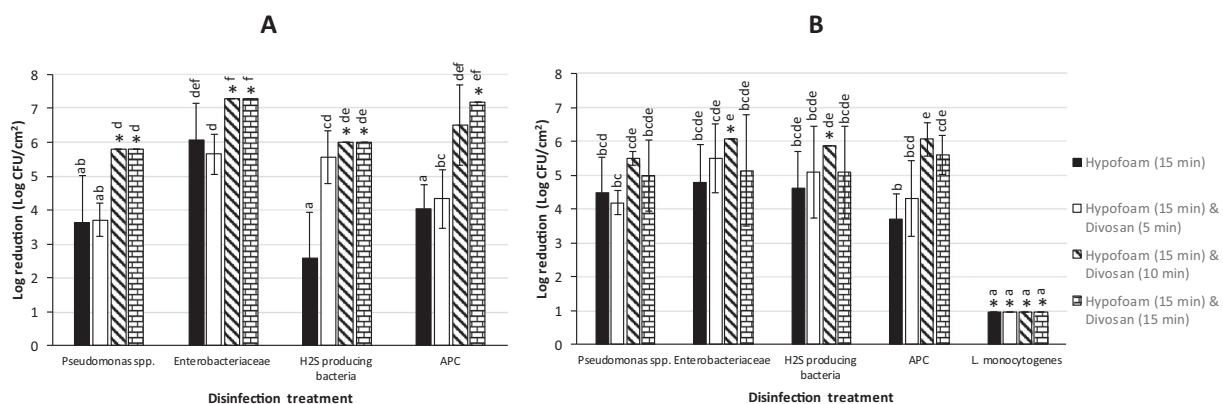


Fig. 7. Log reductions ($\log \text{CFU}/\text{cm}^2$) of *Pseudomonas* spp., Enterobacteriaceae, H_2S producing, APC and *L. monocytogenes* biofilm bacteria, following the four different disinfection treatments. Before disinfection, all bacteria had been left form biofilms, under mixed-culture conditions, on SS coupons incubated at 15 °C, for 168 h (i.e. 7 d), in periodically (i.e. every 48 h) renewable raw FJB (A); or raw FJB also artificially inoculated (at 0 h) with *L. monocytogenes* (at an initial concentration of $2.53 \pm 0.16 \log \text{CFU}/\text{ml}$) (B). The bars represent mean values $\pm$ standard deviations. For each subfigure separately, mean values sharing at least one common lowercase letter shown above the bars are not significantly different at a P value of 0.05. *, counts after disinfection were below the detection limits (DLs) of the plate counting methods (0.95 and $-0.05 \log \text{CFU}/\text{cm}^2$, for spread- and pour-plate methods, respectively); in these cases, the minimal predicted log reductions (based on the DLs and shown in the graphs) were used for the statistical analyses.

4. Discussion

L. monocytogenes is an important foodborne pathogenic bacterium capable of forming biofilms on many surfaces encountered within food processing, surviving probably this way cleaning and disinfection and long term persist (Blackman and Frank, 1996; Martínez-Suarez et al., 2016; Valderrama and Cutter, 2013). Alarming, this has been isolated from seafood and surfaces found in fish processing environments following regular sanitation treatments, together with other microorganisms, mainly of the spoilage and commensal microflora (Rodríguez-López et al., 2015). The composition of such microflora found on seafood processing equipment was indeed the subject of some previous studies. In such a study, results revealed that overall, many microorganisms that are often isolated from fish were also isolated from its processing plants, with some selection depending on processing parameters to be also occurring. Interestingly, the dominant adhering organisms after sanitation were pseudomonads and yeasts independently of the microflora during processing (Bagge-Ravn et al., 2003). In another similar study, the predominant bacteria attached to the processing surfaces in a shrimp factory and a fish processing plant were *Pseudomonas* spp. and Enterobacteriaceae, respectively (Guobjörnsdóttir et al., 2005). Similarly, the microbiota surviving sanitation of conveyor belts in three salmon processing plants was found to be dominated by Gram-negative bacteria and especially *Pseudomonas* spp. (Langsrud et al., 2016).

Noteworthy, surface microflora found in food processing plants may include microorganisms with the potential to increase or decrease colonization and biofilm formation by *L. monocytogenes* (Fox et al., 2014). Of course, the existence of strains which increase such abilities could contribute to the persistence of this pathogen within food industry and consist a potential food safety risk. For instance, Hassan et al. (2004) showed increased attachment of *L. monocytogenes* on wet surfaces precolonized with *P. putida*. In another study, the presence of mixed *Pseudomonas* spp. was found to significantly enhance the colonization of *L. monocytogenes* on SS surfaces (Guobjörnsdóttir et al., 2005). Other spoilage organisms including *Flavobacterium* sp. have been reported to favor biofilm formation by *L. monocytogenes* (Bremer et al., 2001; Jeong and Frank, 1994). EPS production by the resident microflora, could account for the enhanced attachment of *L. monocytogenes*, whose biofilm matrix is not widely known to contain rich amounts of extracellular polysaccharides (Renier et al., 2011). However, in this study, the accumulation of extracellular material was still evident in *L. monocytogenes* mono-species biofilms (Fig. 2), despite the drawbacks SEM surely presents when used to study biofilms (e.g. dehydration of samples, killing of the cells). It should also be noted that the increase in biofilm forming ability of *L. monocytogenes* could also even happen in the sole presence of compounds produced by another species, without the in parallel existence of cells producing them, such as the increase in the number of *L. monocytogenes* EGD-e biofilm cells previously observed in the presence of cell-free supernatant of a *Staphylococcus aureus* strain (Rieu et al., 2008).

Considering the growing evidence on the important influence of both intra- and inter-species interactions on biofilm formation by foodborne pathogenic bacteria, together with their resistance towards disinfection and other stresses (Giaouris et al., 2015; Sanchez-Vizuet et al., 2015), in this study, a cocktail of six *L. monocytogenes* strains of various origins, was left to form biofilms on SS coupons incubated in an aquatic extract of sea bream flesh, for up to 240 h, under both mono-species and mixed-culture conditions. It should be noted that the six different *L. monocytogenes* strains were selected to better represent the genotypic variation potentially present in this versatile pathogenic species, without however having any more information regarding the persistence and/or virulence character of them, whose selection was thus solely based on their different isolation origin. The temperature for incubation (15 °C) was chosen to be close to the thermal conditions encountered in many food production environments. A five-strains

cocktail of four *Pseudomonas* species (*fragi*, *savastanoi*, *putida* and *fluorescens*) or whole fish indigenous microflora, mainly composed of *Pseudomonas* spp., Enterobacteriaceae and H₂S producing bacteria (Table 2), were the ones evaluated for their effect on biofilm formation by the pathogenic cells under the mixed-culture conditions.

In presumable agreement with the dominance of *Pseudomonas* spp. on fish processing surfaces previously described, these are known as strong biofilm forming bacteria (Mann and Wozniak, 2012) and implicated in deterioration of seafood and many other commodities stored aerobically under refrigeration (Tryfinopoulou et al., 2002). Undoubtedly, such characteristics have made them ideal organisms with which to investigate the growth and survival of *L. monocytogenes* within multispecies biofilms (Daneshvar Alavi and Truelstrup Hansen, 2013; Giaouris et al., 2013; Hassan et al., 2004; Lourenço et al., 2011; Puga et al., 2016). Together with *Pseudomonas*, both Enterobacteriaceae and H₂S producing bacteria are known to be implicated in undesirable sensory changes leading to the spoilage and rejection of sea bream and other minimally processed seafood (Boziaris and Parlapani, 2016; Koutsoumanis et al., 1999). In addition, the mixed-species biofilm growth and survival of *L. monocytogenes* with representative members of the Enterobacteriaceae family, namely *Serratia proteamaculans*, *Shewanella baltica*, *Salmonella enterica*, and *Escherichia coli*, have been previously evaluated (Daneshvar Alavi and Truelstrup Hansen, 2013; Kostaki et al., 2012; Rodríguez-López et al., 2017a, 2017b).

Interestingly, *Pseudomonas* spp. bacteria were also here found to clearly dominate in all the mixed-culture sessile communities throughout the whole incubation period (Figs. 1B and 4). In addition, the in parallel sole presence of *L. monocytogenes* cells was found to further increase (ca. 10-fold) their sessile growth (Fig. 1A). In partial agreement with current results, a cocktail of bacterial isolates representing all eight genera isolated from conveyor belts in three salmon processing plants, following sanitation, formed stable biofilms on SS coupons (incubated at 12 °C in salmon broth) of about 10⁹ CFU/cm² after 2 days, while high-throughput sequencing also showed that *L. monocytogenes* represented 0.1–0.01% of the biofilm population and that *Pseudomonas* spp. dominated (Langsrud et al., 2016). With respect to the influence of the presence of the spoilage microflora on biofilm growth by *L. monocytogenes*, two different, but complementary, observations were here made. Thus, when the pathogenic bacteria were left to attach and subsequently form biofilm together with solely the *Pseudomonas* species, the biofilm forming ability of the first ones was either slightly hindered or favored, depending on the incubation day (Fig. 1A). However, the final (240 h) sessile population of *L. monocytogenes* cells did not significantly differ between the two conditions (i.e. mono-/dual-genus), being at both cases equal to ca. 10⁵ CFU/cm² and representing only the 2.3% of the overall sessile cells of the dual-genus community (Fig. 1B). On the contrary, when all the fish indigenous microflora was present from the beginning (0 h), biofilm formation by the pathogen was greatly hampered and never exceeded 3 log CFU/cm² (Fig. 4B). In the future, it would be surely interesting for someone to also analyze this natural background microflora through a high-throughput sequencing technique (Fagerlund et al., 2017; Langsrud et al., 2016), able to provide a more detailed picture of the rich microbiota expected to be present in this niche, than the one derived here through selective plate counting of only the most dominant culturable species. Such info would be subsequently helpful to more accurately determine the potentially developed intercellular interactions regarding their putative impact on biofilm formation and resistance.

Thus, although *Pseudomonas* spp. bacteria were present, at high numbers, at both cases (i.e. under the dual-genus and mixed-culture conditions), in one case, the pathogenic cells managed to reach a rather stable sessile population of ca. 10⁵ CFU/cm², from the 48 h and above (Fig. 1A), while in the latter case, biofilm formation by the pathogen was greatly inhibited, and by the end of incubation (240 h), its sessile population was kept below 10² CFU/cm², that is 1000 times less biofilm

cells compared to the mono-species conditions (Fig. 4B). Quite probably, the underlying reasons behind this inhibition should be the apparent antagonism that should have been occurred between all microorganisms, under such mixed-culture conditions, mainly for nutrients and available space to colonize, including the influence of such and possibly other developed competitive relationships (e.g. production of antimicrobials) on the planktonic growth of each strain (Rendueles and Ghigo, 2012). In agreement with this observation, other previous studies on biofilm formation by *L. monocytogenes* under mixed-culture conditions with other bacteria have shown similar inhibitions. For instance, biofilm formation by *L. monocytogenes* was reduced in co-cultures with *P. fragi* (Norwood and Gilmour, 2001) and other various Gram-negative bacteria, including *Serratia* spp., *Aeromonas* sp. and *P. fluorescens* (Gudmundsdóttir et al., 2005; Daneshvar Alavi and Truelstrup Hansen, 2013), but also with Gram-positive ones, including *Enterococcus* spp. and other lactic acid bacteria (da Silva Fernandes et al., 2015; Leriche et al., 1999; Minei et al., 2008; Zhao et al., 2004).

All these results surely highlight the complexity mixed-culture biofilms present, together the inability for someone to draw general conclusions on the influence of interspecies interactions between spoilage and pathogenic bacteria on biofilm formation by the latter, but also in general. Thus, the biofilm forming behavior of a species upon co-culture with other species cannot be predicted, since this seems to depend on many parameters, such as the other species employed, the culture and setup conditions used for biofilm formation (including the availability of nutrients, incubation time, sequence of attachment, static or flow conditions), and surface used to support growth (Giaouris et al., 2015). In addition, results gained during numerous studies on bacterial attachment and biofilm formation indicate that factors involved in this process can be strain specific and that caution should be taken when trying to extrapolate results obtained employing only one strain (Borucki et al., 2003; Mosquera-Fernández et al., 2014; Nilsson et al., 2011). That is why in the current study multi-strain *L. monocytogenes* and *Pseudomonas* cocktails were always employed. To this direction, previous studies have shown that the different *L. monocytogenes* strains being present in mixed-culture biofilm communities (both mono- and mixed-species) may not contribute at the same level either to their formation or to their antimicrobial resistance, with some of them to manage to dominate over the others, depending on the surrounding conditions (Giaouris et al., 2013; Kostaki et al., 2012). However, other studies did not reveal any difference at these two important phenotypic properties (which are also potentially linked to persistence) upon comparing *L. monocytogenes* strains presenting individual genetic traits and in parallel differing in their abilities to persist in food production facilities (Fagerlund et al., 2017; Overney et al., 2017).

It should also be noted that under the mixed-culture conditions, pathogenic cells were, in purpose, left here to attach to SS coupons together with all the fish indigenous microflora from the beginning, without the existence of any prior step (where *L. monocytogenes* might have been left attach alone), and subsequently altogether were left to form mixed-culture biofilm (Section 2.3.3). Undoubtedly, this direct incubation of the sterile SS coupons in the raw FJB also inoculated with the pathogenic bacteria (ca. 10^{2-3} CFU/ml), should better simulate an unwanted scenario that might be encountered in the fish industry. Noteworthy, it is also known that just the sequence of initial colonization of a surface can still influence the population's composition, structure and physiology of a mixed-culture sessile community (Wang et al., 2015). To simulate real fish processing conditions, a model fish juice substrate was also here used to support biofilm development. It is additionally known that in any food processing environment, food residues adsorbed on a substratum create conditioning films that can affect, either enhance or reduce, bacterial attachment and subsequent biofilm formation (Bernbom et al., 2009, 2011; Brown et al., 2014; Vejborg et al., 2008; Whitehead et al., 2010). Moreover, the presence of food residues may increase the resistance of surface-adherent bacteria to disinfection, rendering sanitization processes ineffective and

encouraging cross contamination (Gram et al., 2007).

In the current study, the biocide tolerance of the developed mono-species and mixed-culture sessile communities against two common food industry disinfectants (Hypofeam, Divosan) was also evaluated. Hypofeam VF6 is a highly chlorinated, combined cleaner and disinfection product, containing a high level of hypochlorite and a blend of chelating agent and surfactant/wetting agents. Divosan SU 319/VT8w is a general-purpose terminal disinfectant for open plant applications based on quaternary ammonium compound (QAC). Both are designed for daily use in the food, beverage and dairy industries. However, all here tested disinfection treatments, gradually applied for 15 to 30 min, were insufficient against *L. monocytogenes* mono-species biofilm communities, with the resistance of the latter to significantly increase from the 3rd to 7th day of incubation (Fig. 5). This higher tolerance of *L. monocytogenes* in single-species biofilms over time against both peracetic acid and QAC disinfectants has been also previously shown (Fagerlund et al., 2017; Saa Ibusquiza et al., 2011). On the other hand, it is reassuring that the combined application of both disinfectants each one for 15 min (in line with the daily practice of the aquaculture company) always resulted in undetectable viable bacteria on the SS surfaces previously incubated in solely raw FJB for 7 d (168 h) (Fig. 7A). However, the same treatment was insufficient to kill completely the bacteria previously left to form mixed-culture biofilms together with the pathogenic cells, still presenting log reductions of ≥ 5 log CFU/cm² against all the microbial groups here tested (Fig. 7B). Hence, the incorporation of *L. monocytogenes* cells in the mixed-culture sessile community developed by the fish indigenous microflora seems to alter its overall resistance against disinfection, which is especially interesting considering that these cells: (i) consisted only the ca. 0.0002% of the total sessile APC before disinfection (Table 2), and (ii) were not at all recovered following even the mildest treatment (application of Hypofeam for 15 min). This last is quite probably associated their low number being present before disinfection. However, it is still worthy that the rest rich surface microflora being present before disinfection together with the pathogenic cells (APC > 7.50 log CFU/cm²) was not able to protect the relatively low biofilm population of the latter. Thus, no protection of pathogenic cells by the spoilage surface microflora could be here traced, as shown in some other previous studies (Puga et al., 2016; van der Veen and Abee, 2011).

In conclusion, the current study increases our current limited knowledge on both the population dynamics and resistance of *L. monocytogenes* biofilm cells upon exposure to disinfection, under conditions resembling those encountered within the seafood industry and should thus be considered upon designing and applying effective anti-biofilm strategies in such environments. Undoubtedly, such knowledge could more generally provide the information necessary to modify processes in food processing environments to reduce unwanted *L. monocytogenes* persistence and therefore reduce the contamination of food products, obviously with potential important impacts to food safety and public health.

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